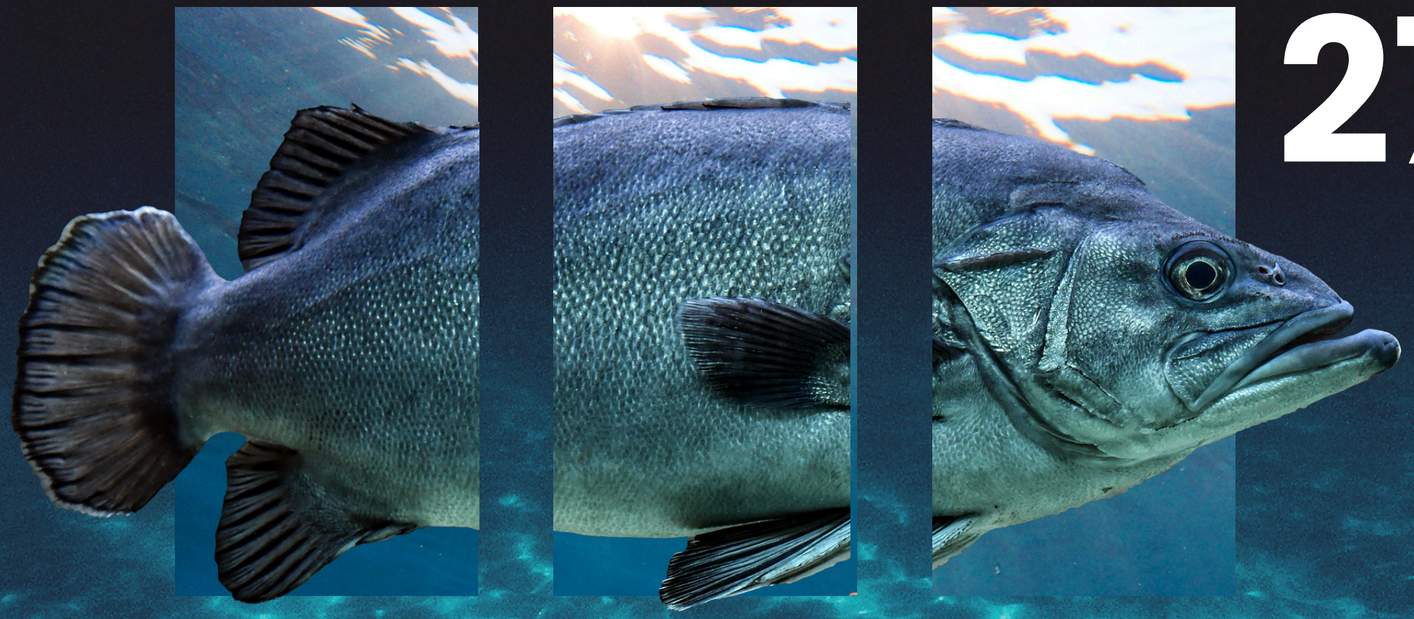
BLUE SHIFT

UNDERSTANDING THE RELATIONSHIP BETWEEN CHANGING OCEAN CONDITIONS AND PHILIPPINE FISHERIES PRODUCTION

THE PEOPLE WHO FEED THE NATION FACE A SEA THAT KEEPS CHANGING

An island nation that depends heavily on fisheries for food, livelihoods and economic activity



27.4%

of fisherfolk live in poverty, almost double the national rate

Ocean conditions keep changing: rising sea temperatures and shifting productivity can influence fishing conditions

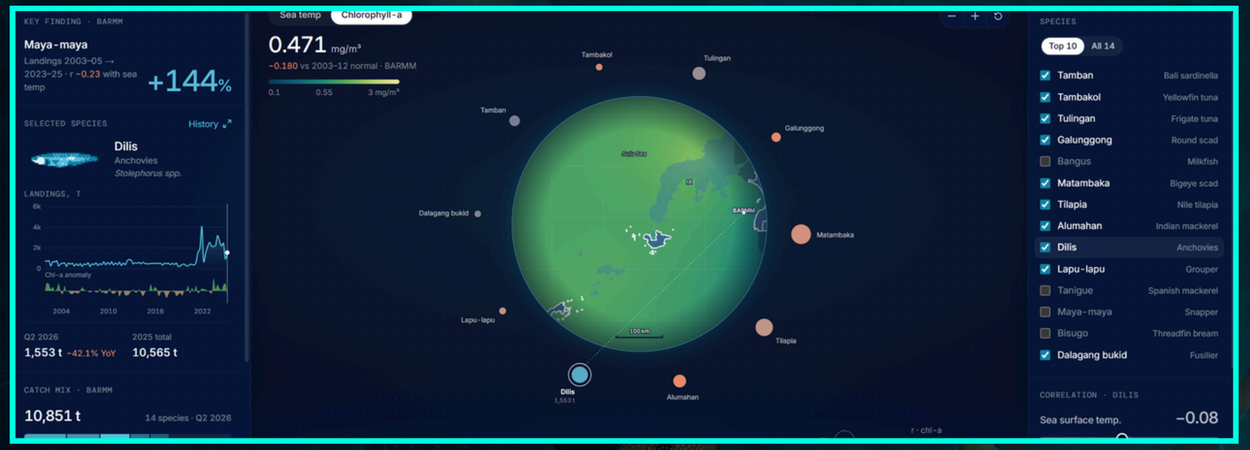
But satellite ocean data and fisheries statistics are often examined separately

HOW ARE CHANGING OCEAN CONDITIONS ASSOCIATED WITH FISHERIES PRODUCTION?

INITIAL STUDY AREA: MINDANAO

- Built on available NASA satellite observations and compatible PSA fisheries statistics
- BARMM and Zamboanga Peninsula land about 4 in 10 tonnes of Philippine fish
- In SOCCSKSARGEN, small-scale marine catch fell by two-thirds in 2025
- A starting point before expanding to other Philippine fishing regions

BLUESHIFT



- **Explore ocean conditions:** 24 years of sea temperature and ocean colour
- **Track local fisheries production:** 24 provinces, 66 species, every quarter
- **Investigate** how environmental variables relate to changes in fish production over time

24 YEARS **24** PROVINCES **66** SPECIES IN ONE PLACE

HOW IT WORKS

FOUR STEPS ON ONE TIMELINE: 96 QUARTERS, JUL 2002 TO JUN 2026

96 QUARTERS
OF SEA AND CATCH, SIDE BY SIDE

1

DATA INPUT • SEA

MONITOR THE OCEANS

NASA sea-surface temperature (SST) and chlorophyll-a levels.

NASA MODIS / VIIRS — **SST + CHL-A**

2

DATA INPUT • CATCH

TRACK LOCAL FISHERIES

PSA catch volume across regions, provinces, and species.

PSA OPENSTAT — **VOLUME & SPECIES**

3

SYNTHESIS • ANALYSIS

EXPLORE RELATIONSHIPS

Statistical correlations pairing ocean shifts with catch.

CROSS-CORRELATION — **SEA + CATCH**

4

OUTPUT • SPACIAL

DISCOVER REGIONAL INSIGHTS

Spatial and time patterns across 24 Mindanao provinces.

MINDANAO COAST 24 PROVINCES

JUL 2002 (Q3) — **UNIFIED 24-YEAR TIME SERIES . 96 SYNCHRONIZED QUARTERS** — **JUN 2026 (Q2)**

EXPECTED IMPACT

MAKING ENVIRONMENTAL AND FISHERIES DATA ACCESSIBLE SO THREE CORE GROUPS CAN ACT ON CHANGING OCEAN CONDITIONS:

RESEARCHERS

Investigate ocean-to-catch correlations across 96 quarters without manual data wrangling.

FISHERIES MANAGERS

Anticipate provincial harvest shifts using plain-language quarterly sea advisories.

POLICYMAKERS

Ground coastal resilience programs and LGU support in 24 years of empirical evidence.

MEET THE TEAM

TEAM LEAD

IAN JURE MACALISANG

DOCUMENTATION LEAD

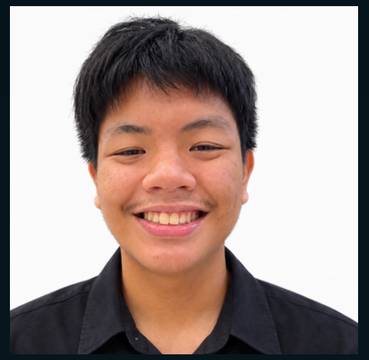
ROGELIO MANDAMIAN III

RESEARCH LEAD

KYLE BAUTRO

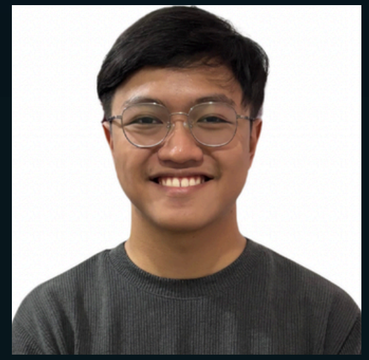
DESIGN LEAD

GABRIEL MAR RAMIREZ



TECHNICAL LEAD

HANS DEL MUNDO



SYSTEMS ENGINEERING LEAD

KENT AUXTERO

BlueShift empowers users to become Earth System Trend Detectives by turning environmental and fisheries data into insights that can support informed decision-making.